

Mock Exam Questions

AI Safety Course at the University of Tübingen

Instructions. Answer all questions in the boxes provided. Conceptual questions expect up to 4 sentences. Each question is worth an equal number of points (1 point per question).

Part I. Adversarial ML, Jailbreaks & Prompt Injections

Q1. What does “the attacker moves second” in AI security mean, and why does it make static benchmark numbers unreliable?

Answer:

Part II. Open-Weight LLM Safety

Q2. *Abliteration* is the technique that strips a model’s ability to refuse by removing the direction in its activations that encodes refusal. State (a) what is removed, (b) the linear-algebra operation used, and (c) whether any training is required.

Answer:

Part III. Transparency: Detection & Watermarking

Q3. The Binoculars detector uses two models, an “observer” and a “performer”. Explain why it compares two perplexities instead of thresholding one.

Answer:

Part IV. Data: Privacy, Memorization & Data Integrity

Q4. State the three privacy threat models (extraction, reconstruction, inference) and describe the differences and risks.

Answer:

Part V. Alignment Tools

Q5. Direct Preference Optimization (DPO) is summarized as “your language model is secretly a reward model.” What two expensive components of RLHF does DPO eliminate?

Answer:

Part VI. LLM Agents

Q6. Explain *progressive disclosure* in Agent Skills, and why it lets a skill’s effective size be essentially unbounded relative to the context window.

Answer:

Part VII. Agent Security

Q7. State the three capabilities of the “lethal trifecta” that enable harmful prompt injections. Explain why it is the combination, not any single capability, that makes an agent dangerous.

Answer:

Part VIII. Scheming & Deception

Q8. State the distinction between deception and scheming, and give the definition of scheming.

Answer:

Answer Key

Part I — Adversarial ML, Jailbreaks & Prompt Injections

- Q1. The defender publishes a fixed defense first; the attacker then reads it and builds an *adaptive* attack tailored to that specific defense. Static benchmark numbers are gathered against a fixed, pre-specified set of attacks — before any such adaptive attack exists — so they only measure robustness against the attacks already tried. That is an upper bound, and it routinely collapses (often to near 0%) once an adaptive attacker responds, which is why the reported number overstates real robustness.

Part II — Open-Weight LLM Safety

- Q2. (a) The single refusal direction $\hat{r}$ in the residual stream; (b) directional ablation — projecting that direction out of the weights/activations; (c) no training — it is a training-free weight edit.

Part III — Transparency: Detection & Watermarking

- Q3. Neither model's perplexity alone separates human from machine; their *ratio* cancels prompt- and topic-driven variation, so a genuinely model-written passage shows anomalously low relative perplexity while human text does not.

Part IV — Data: Privacy, Memorization & Data Integrity

- Q4. **Extraction:** with weights + the right prompt, pull verbatim private data out of the model. **Reconstruction:** given partial context about a person, fill in a hidden attribute. **Inference:** from arbitrary public text a person wrote, predict hidden attributes. Inference is worst because it needs *no* memorization and no presence in training data — so deduplication, unlearning, and extraction defenses do nothing against it; any capable model is a privacy risk to anyone who writes publicly.

Part V — Alignment Tools

- Q5. (1) The separately trained reward model, and (2) sampling from the policy inside an RL loop. DPO turns preference learning into a single supervised classification loss on preference pairs — just gradient descent, no RL.

Part VI — LLM Agents

- Q6. Progressive disclosure loads a skill **in levels**: only the **name + description** are preloaded, the **full SKILL.md body** loads only when the agent judges the skill relevant, and **bundled files/scripts** are read or run only as needed. Because the agent pays context **only for what it opens**, the total material a skill carries can far exceed the context window.

Part VII — Agent Security

- Q7. (1) **Access to private data**, (2) **exposure to untrusted content**, and (3) **the ability to communicate externally** (an exfiltration channel: send mail, fetch a URL, post a message). Each is individually useful; only when **all three are present at once** can the agent become a data-theft machine, so the danger is in the combination.

Part VIII — Scheming & Deception

- Q8.** A one-off lie is deception. Scheming is the narrower case: deception used strategically to pursue a hidden goal (including during training). Scheming = covertly pursuing mis-aligned goals while hiding one's true capabilities and objectives.